Ex:No: 1D Date: 04.02.26	Password Strength Analysis Using String Operations

Problem Description:

An online system requires users to create secure passwords. An R program is needed to analyze a user-entered password by checking its length, converting case, and identifying whether it contains numbers and special characters using string operations.

Objective:

To demonstrate the application of string manipulation functions in R for analyzing and validating password strength.

Sample Input	Expected Output
Enter password: Pass@123	Password Length: 8 Uppercase Version: PASS@123 Contains Number: TRUE Contains Special Character: TRUE
Enter password: Pass	Password Length: 4 Uppercase Version: PASS Contains Number: FALSE Contains Special Character: FALSE

Program:

```
# Password strength analysis using strings in R
# User input password
password <- readline("Enter password: ")

# Length of password
password_length <- nchar(password)

# Case conversion
uppercase_password <- toupper(password)

# Check for number
contains_number <- grepl("[0-9]", password)
```

Input:

1) Enter password: PassWord!

Output:

Password Length: 9
Uppercase Version: PASSWORD
Contains Number: FALSE
Contains special characters: TRUE

2) Enter password: helloworld

Password Length: 10
Uppercase Version: HELLOWORLD
Contains Number: FALSE
Contains Special character: FALSE

Check for special character

```
contains special <- grepl("[^A-Za-z0-9]", password)
cat("Password Length: ", password_length, "\n")
cat("Uppercase version: ", uppercase_password, "\n")
cat("contains Number: ", contains_number, "\n")
cat("Contains special character: ", contains_special, "\n
```

Result:

Thus the program analyzes the password using string functions in R to determine its length and character composition. It demonstrates how string operations help in basic password validation.